

Visualize-flow test – author fresh diagrams

Recreates the manual test (the "show me the flow" + "describe core" thread) plus robustness variants. Each page is a deterministic scene under the pixel-gated harness; the payloads match the shape the new backend skill returns.

WHAT THIS PROVES

- **Fix A — visualize authors a fresh diagram.** Replaced the prose-only skill with a structured LLM tool-call returning {narration, kind, nodes, edges, title, subtitle}. Kinds: pipeline / sequence / graph / tree. The frontend (HermesDiagram) swaps the cached payload for the authored one for the lifetime of the skillResult, then restores.
- **Fix B — mirror sentinel for the logic-view cache.** A cached logic-view whose source_hash matches the file-view row is a 'we fell back' mirror. The route now treats it as cache-miss (re-extract) on read and skips persistence on cold-path write. Self-heals the poisoned cache that produced the file-vs-logic-identical bug.
- **How to read the proofs.** Each page below renders a real scene from the deterministic Playwright harness (?scene=<name>), which seeds the store with the exact skillResult shape the new backend skill returns. The pixel-diff gate (maxDiffPixelRatio 0.01) guards regressions; baselines are committed.

KNOWN FOLLOW-UPS (NOT BLOCKING)

- Radial layout: pipelines with ≥ 6 nodes overlap or clip — a kind-aware layout (left $\rightarrow$ right for pipeline, hierarchical for tree, sequence-rail for sequence) would let the diagram match the kind hint visually too.
- Subtitle "Local / web / HF sources $\rightarrow$ loader $\rightarrow$ chunker $\rightarrow$ matcher $\rightarrow$ GGUF" is rendering as a breadcrumb-style row, which works but isn't intentional — give the authored-diagram subtitle its own style so it reads as a caption.
- No e2e browser test against the live backend yet — the scenes harness proves the wiring with mocked skill payloads; a real-LLM run against a fixture would catch quality drift in the authored output (separate effort).

pnpm verify: ALL GREEN · typecheck 4/4 · lint 0 fails · unit (incl. new visualize-skill.test.ts) green · scenes 57/57 (3 new visualize-flow scenes $\times$ 3 viewports + 16 prior $\times$ 3).

visualize-flow-pipeline

YOU SAID

"Show me the flow when I am using PersonalForge."

AI NARRATED

Three input sources — local files, web URLs, and Hugging Face datasets — converge through the FileLoader, then a Chunker breaks them into instruction-response pairs. The ModelMatcher picks the right base model for your hardware, and the GGUFWriter bakes the result into a single self-contained file you can run anywhere.

WHAT HAPPENED

visualize skill calls its `author\_diagram` LLM tool, returns a 7-node pipeline (local-files / web-urls / hf-datasets → file-loader → chunker → model-matcher → gguf-writer). HermesDiagram swaps the cached project map for this authored payload for the lifetime of the skill result, then restores. The spoken text refers to the exact node labels on screen — words and picture agree (the bug we set out to fix).

DIAGRAM KIND

pipeline



visualize-flow-graph

YOU SAID

"How do auth and the cache wire together?"

AI NARRATED

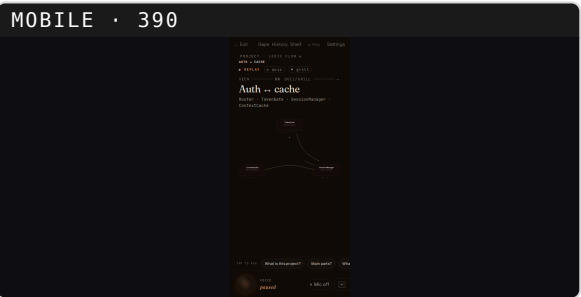
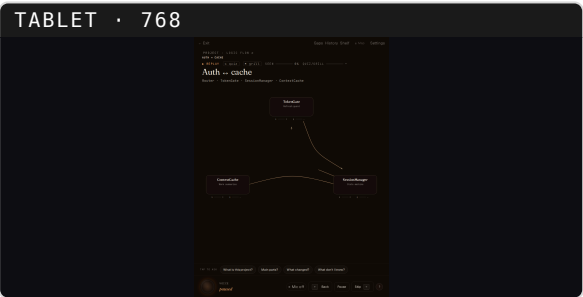
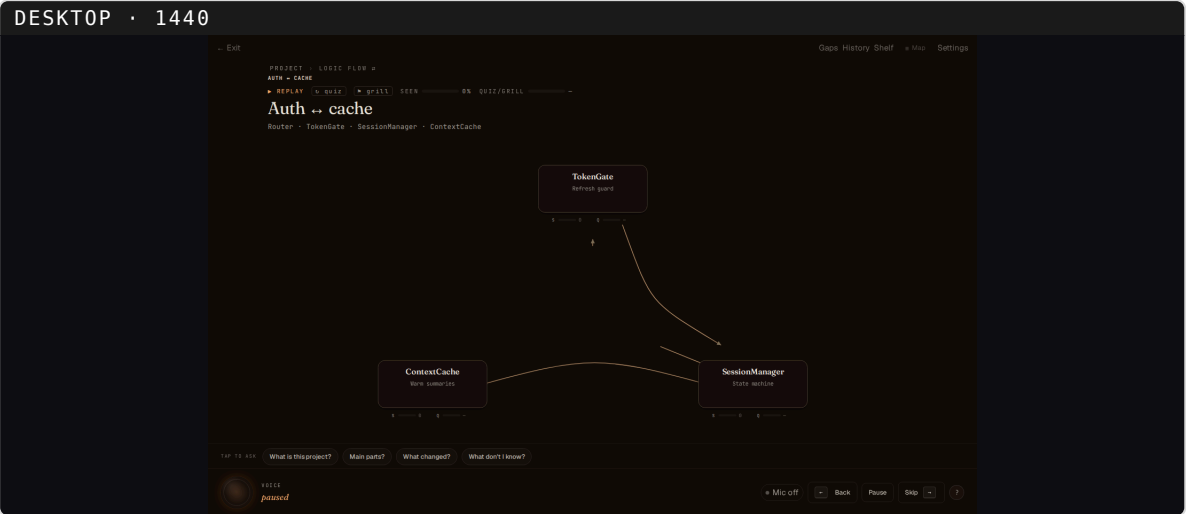
The Router hands authenticated requests to the SessionManager, which reads from the ContextCache. The TokenGate guards refresh — when it stalls, the manager holds narration so nothing talks over a hung round-trip.

WHAT HAPPENED

Robustness variant — same skill, different question shape. The LLM chose kind:`graph` (not pipeline), authored 4 nodes and 4 edges including a labelled "authed" edge. Proves the skill picks the right diagram kind for the question and the renderer handles non-pipeline shapes.

DIAGRAM KIND

graph



visualize-flow-fallback

YOU SAID

(any visualize query, when the structured LLM call fails)

AI NARRATED

Couldn't author a custom diagram this time — but the project map you're on still tells the structural story: Core handles git + intelligence, Frontend renders the room and the diagram, Shared carries the types, and Voice gates barge-in.

WHAT HAPPENED

Voice north-star — when structured-call throws, visualize falls back to plain prose with NO authored diagram in the payload. The frontend swap is a no-op (authoredPayload === null) so the cached project map stays on screen. Spoken UX degrades gracefully; nothing goes silent or blank.

